

Bundling Method of Periodically Generated Packets for Collision-Free Transmission by Uplink OFDMA in Wireless Sensor Networks

Yuya Onogawa, Yosuke Tanigawa, Hideki Tode

*Department of Core Informatics, Graduate School of Informatics, Osaka Metropolitan University
Sakai, Osaka 599-8531, Japan*

Email: {sb22764x@st., tanigawa@, tode@}omu.ac.jp

Abstract—In the recent Internet of Things (IoT) context, smart IoT environments such as smart factory ones must transfer periodically generated data packets as well as aperiodically generated ones with small loss rate. With Orthogonal Frequency-Division Multiple Access (OFDMA) transmission introduced in IEEE 802.11ax, multiple packets are transferred efficiently without collision. In this paper, focusing on the periodicity of the generation of periodic packets, we propose a method that repeatedly transfers periodic packets simultaneously by bundling them into OFDMA transmissions. Packets that are transferred simultaneously by an OFDMA transmission are determined so that total channel occupancy period is minimized based on mathematical analysis. Minimizing the channel occupancy period also contributes to alleviating collision among aperiodic packets because time ratio for transmitting them is increased. Through performance evaluation with computer simulation, we demonstrate the effectiveness of the proposed method.

Index Terms—OFDMA, IEEE 802.11ax, Smart Factory, Smart IoT, Wireless Sensor Networks

I. INTRODUCTION

In smart Internet of Things (IoT) environments such as smart factory ones, periodically generated data packets as well as aperiodically generated ones must be transferred with low loss rate to satisfy various requirements like permissible transfer delays [1], and wireless sensor networks (WSNs), used to transfer such packets, need to be operated at low costs. WSNs based on the IEEE 802.11 standard [2] can be used at low costs because its random channel access control needs low computational complexity and it is widely diffused as Wi-Fi. However, the random channel access control with Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) causes packet collision, and as results, transmission delay fluctuation and frequent packet loss occur.

Orthogonal Frequency-Division Multiple Access (OFDMA) transmission is newly introduced in the IEEE 802.11ax standard [3] [4] in 2021, and can be used to transfer multiple packets simultaneously without collision. In OFDMA transmission, a wireless channel is divided into several resource units (RUs), and multiple packets are transferred simultaneously with the

RUs. In uplink OFDMA transmission for packet transmission from sensor nodes to their sink, which is the dominant direction for WSNs generating periodic packets from sensing devices, how to allocate RUs to sensor nodes is classified into scheduled access (SA) and random access (RA). In SA, the RU allocation to source nodes is decided by the sink based on the their transmission buffer statuses, which results in no packet collision. However, each source node must notify the buffer status in advance by using buffer status report (BSR). On the other hand, in RA, each source node randomly selects the RU to be used for its packet transmission based on the information provided about the available RUs. Although RA does not require the notification of the BSR, packet collision occurs when multiple source nodes select a same RU. To achieve low-loss packet transmission regardless of the packet type, SA OFDMA seems appropriate. Hence, we focus on SA OFDMA.

In this paper, focusing on the periodicity of the generation of periodic packets, we propose a method that bundles periodically generated packets into uplink OFDMA transmissions for collision-free transmission. Based on the periodic packet generation intervals at the source nodes, packets that are transferred simultaneously by being bundled into an OFDMA transmission can be determined in a planned and systematic way. In the proposed method, periodically generated packets are bundled into OFDMA transmissions so that total channel occupancy period is minimized based on mathematical analysis. Minimizing the channel occupancy period for OFDMA transmissions also contributes to alleviating collision among aperiodic packets because time ratio for transmitting them is increased.

II. RELATED WORKS

In recent studies, OFDMA transmission is applied for improving data transfer throughput and delay, channel utilization efficiency, and so on. Chen et al. [5] propose scheduling channel access with TWT (SCAT) in which each wireless station (STA) is assigned the optimal beacon interval to activate RA OFDMA transmissions from one of the beacon intervals multiplied by a power of two based on the packet arrival status, with the objective of maximizing the packet transfer throughput from STAs to base station per target wake

This work includes results of the project entitled “R&D on Technologies for Adaptive Sharing of Radio Resources for Realtime Applications,” which is supported by the Ministry of Internal Affairs and Communications as part of the research program “R&D for Expansion of Radio Wave Resources (JPJ000254)”.

time (TWT) session cycle in a wireless LAN (WLAN). A similar approach is considered also for overlapping WLANs [6]. However, these methods do not consider the periodicity of generation timings of periodic packets, and assume RA OFDMA in contrast to our proposed method that focuses on SA OFDMA. Scheduled transmission opportunities (S-TXOP) [7] [8] facilitates periodic uplink and downlink OFDMA transmissions, taking into account the characteristics of periodically generated packets. It aims to provide low-latency SA OFDMA transmission opportunities by minimizing RU allocation overhead through quasi-static RU allocation and reducing control frame overhead. However, it assumes IEEE 802.11be, the next standard after IEEE 802.11ax, and time synchronization between base station and periodic packet generating nodes for quasi-static RU scheduling of periodic packet flow. The time synchronization should be avoided for WSNs because the computational and implementation costs of sensor nodes are increased. Schneider et al. [9] propose a scheme that provides deterministic channel access using OFDMA transmission by restricting channel access based on the enhanced distributed channel access (EDCA) through the extended multi-user (MU) EDCA mechanism in the 6 GHz band without the constraints of backward compatibility. However, none of these studies consider the periodicity of generation timings of periodic packets, and control their transmission timings so that they are repeatedly transmitted simultaneously by OFDMA.

III. PROPOSED METHOD

In the proposed method, packets periodically generated at source nodes are transferred by being bundled into OFDMA transmissions so that total channel occupancy period is minimized by taking the periodicity into account. This paper assumes that each source node generates packets either periodically at a constant interval or aperiodically, and transmits them to the sink. All the packets generated at a source node have a same size but the sizes may be different among different source nodes. *OFDMA transmission executable timings* where uplink OFDMA transmissions can be triggered by the sink are set at a constant interval T_{ofdma} on the time axis. When a periodic packet is received at the sink from a new source node that periodically generates packets, the OFDMA transmission timings from the source node are selected from among OFDMA transmission executable timings at a constant interval equal to the packet generation interval. By transmitting multiple periodic packets by OFDMA, collision among the packets is avoided. Minimizing the channel occupancy period for OFDMA transmissions also contributes to alleviating collision among aperiodic packets because time ratio for transmitting them is increased.

A. Bundling Periodic Packets into OFDMA Transmissions

When the first periodic packet from a new source node n_j , whose packet generation interval is T_j , is received at the sink, OFDMA transmission timings of the following periodic packets from the source node n_j are selected from among OFDMA transmission executable timings. From the reception

time of the first periodic packet, a prescribed period called *common period* for the source node n_j is considered on the time axis. The length T of the common period is the least common multiple of the packet generation intervals at all the active source nodes that periodically generate packets. The common period is divided into smaller periods of T_j whose number is T/T_j . In each period of T_j , T_j/T_{ofdma} OFDMA transmission executable timings exist under the assumption that T_j is divisible by T_{ofdma} . Figure 1 shows the relationship of the common period, the period of T_j , and OFDMA transmission executable timings in case of $T/T_j = 2$ and $T_j/T_{ofdma} = 4$. For the k th ($k = 1, 2, \dots, T_j/T_{ofdma}$) OFDMA transmission executable timing in the m th ($m = 1, 2, \dots, T/T_j$) period of T_j in each common period, the increase in the channel occupancy period when allocating the timing to packet transmission from the source node n_j is calculated by

$$I_{m,k} = \begin{cases} D_{\text{overhead}}^{OFDMA} + D_{\{n_j\}}^{OFDMA}, & \text{when } |N_{m,k}| = 0 \\ D_{\{n_j\}}^{Control} + D_{N_{m,k} \cap \{n_j\}}^{OFDMA} - D_{N_{m,k}}^{OFDMA}, & \text{when } |N_{m,k}| \neq 0. \end{cases} \quad (1)$$

$D_{\text{overhead}}^{OFDMA}$ is the overhead period that consists of durations required to transfer a trigger frame (TF) and a multi-station block ACK (MSTA-BA) and SIFS durations after the TF and before the MSTA-BA. $D_{\{n_j\}}^{Control}$ is the control overhead period caused by the additional information for RU allocation and acknowledgment required for adding node n_j to the OFDMA transmission, such as the UserInfoField of the TF and the BA information of the MSTA BA. $N_{m,k}$ is the set of source nodes whose periodic packet transmission opportunities by OFDMA are already allocated to the corresponding OFDMA executable timing. D_N^{OFDMA} is the duration required to transmit packets from the source nodes in the set N to the sink simultaneously by OFDMA, and is calculated as

$$D_N^{OFDMA} = \frac{B_{max}}{V_{|N|RU}}. \quad (2)$$

B_{max} and $V_{|N|RU}$ are the maximum size of packets from the source nodes in the set N and the OFDMA transmission rate from the source nodes, respectively. From the above, the additional channel occupancy period per common period when allocating the source node n_j to the k th OFDMA transmission executable timing in each period of T_j is calculated as

$$I_k = \sum_{m=1}^{T/T_j} I_{m,k}. \quad (3)$$

Finally, the k th OFDMA executable timings with the minimum I_k are selected and allocated to the new source node n_j per common period.

B. Packet Transmission from Source Nodes

Periodically generated packets are transferred by uplink OFDMA transmissions invoked by the sink according to the OFDMA transmission schedule allocated to the source nodes

based on the procedure in Sec. III-A. At each OFDMA transmission executable timing, the sink triggers OFDMA transmission by transmitting a TF after waiting for the DIFS or AIFS period and the following backoff period. RUs are allocated to the source nodes whose packet transmission is allocated at the timing. Although the proposed method allocates RUs to the source nodes without using BSR, with BSR polling (BSRP) and one handshake overhead, it is possible to allocate RUs to the source nodes using BSR.

In contrast, aperiodically generated packets are transmitted based on CSMA/CA. Each source node that aperiodically generates packets obtains transmission opportunities autonomously based on CSMA/CA and transmits their packets.

IV. PERFORMANCE EVALUATION

We implemented the proposed method on QualNet v9.0 [10], and evaluated its performance by the computer simulation. Section IV-A explains the simulation scenario and Sec. IV-B shows simulation results.

A. Simulation Scenario

A sink is located at the center of a circle with a radius of 20 m, and each source node is randomly placed within the circle area. The center frequency and the bandwidth of the wireless channel are 2437 MHz and 20 MHz, respectively. The sink and all the source nodes use IEEE 802.11ax physical layer and IEEE 802.11e [11] MAC layer. Modulation and coding scheme (MCS) index is set to 2 whose transmission rate is 25.8 Mbps. In the uplink OFDMA transmission, MU-RTS/CTS handshake is used before transmitting TF. The maximum OFDMA transmission duration is set to 4.095 ms. The OFDMA transmission interval T_{ofdma} is set to 10 ms. We run the simulation 20 times for each number of source nodes, and each simulation duration is 300 s.

Each source node generates packets periodically or aperiodically, and transmits them directly to the sink. Source nodes that periodically generate packets and source nodes that aperiodically generate packets exist in the ratio of 3:2. The source nodes that periodically generate packets are further divided into two types based on the assumed applications, with a ratio of 1:1. For the former type, the generation interval and the size are set to 100 ms and 1200 bytes, respectively. For the latter one, the generation interval and the size are set to 100 ms and 200 bytes, respectively. For the aperiodic packets, 3 packets of 500 bytes are generated at intervals that follow an exponential distribution with an average of 0.05 s. These settings assume applications in smart factory use cases [1]. The applications have large delay time tolerances, and all the packets can be treated as non-delay-bounded ones. As prioritized channel access control with IEEE 802.11e, AC_VO is used to transmit TFs for triggering the uplink OFDMA transmission, and AC_BK is used for the other types of frames. Aperiodic frames are transmitted based on CSMA/CA. If a packet collision occurs, the collided packets are not retransmitted.

In this evaluation, the following methods are compared. First, the proposed method, which is referred to as “Proposed method with MinTxPeriod,” implements all the functions explained in Sec. III. Second, the simplified proposed method, referred to as “Proposed method with First-Fit,” implements simplified allocation of OFDMA transmissions to periodic packets based on a first-fit manner in which transmission timings of periodically generated packets of a new source node by OFDMA are selected from the closest OFDMA transmission executable timings from the time of its first periodic packet arrival. Third, IEEE 802.11 Distributed Coordination Function (DCF) transmits all packets based on CSMA/CA with AC_BK as a basic benchmark, and is referred to as “DCF with AC_BK.” Fourth, as a benchmark with OFDMA transmission, in the basic OFDMA method referred to as “Standard OFDMA,” uplink OFDMA transmissions are executed for transferring periodically and aperiodically generated packets at a constant interval T_{ofdma} . In each OFDMA transmission, if the number of source nodes that buffer packets is greater than the number of RUs, source nodes equal to the number of RUs are randomly selected from among all the source nodes buffering packets and the RUs are assigned to them.

B. Simulation Results

The results of the packet loss rate and the end-to-end packet transfer delay are shown in Figs. 2 and 3, respectively. The packet loss rate is the ratio of packets that do not reach the sink to all the generated packets. The packet transfer delay is average delay of packets from the generation at source nodes to the arrival at the sink. For the transfer delay of periodically generated packet, the delay excluding the waiting time (WT) at source node from the generation time to the time at which its OFDMA transmission is started is also evaluated as the legend “Periodic w/o WT” in Fig. 3. In Fig. 2, the loss rates of periodic packets in the proposed methods with MinTxPeriod and First-Fit almost overlap below 10^{-3} , and the other results are above 10^{-3} .

From Figs. 2 and 3, both the proposed method with MinTxPeriod and the proposed method with First-Fit greatly suppress the loss rate of periodic packets compared to DCF with AC_BK and Standard OFDMA. This is because periodically generated packets are transferred with OFDMA transmissions, which are executed periodically, in parallel without collision. Furthermore, the proposed method with MinTxPeriod also suppresses the loss rate of aperiodic packets up to 36% compared to the proposed method with First-Fit because it minimizes the increase in the channel occupancy period due to the uplink OFDMA transmission, and time ratio used for transmitting aperiodic packets is increased. Although the end-to-end packet transfer delays of the proposed methods with MinTxPeriod and First-Fit become larger than both Standard OFDMA and DCF with AC_BK, it does not matter because these values are below the larger and negligible delay constraints, such as 500 ms or a few seconds. The delays excluding the WT are much smaller than the end-to-end packet transfer delays in the proposed methods. This means that, by

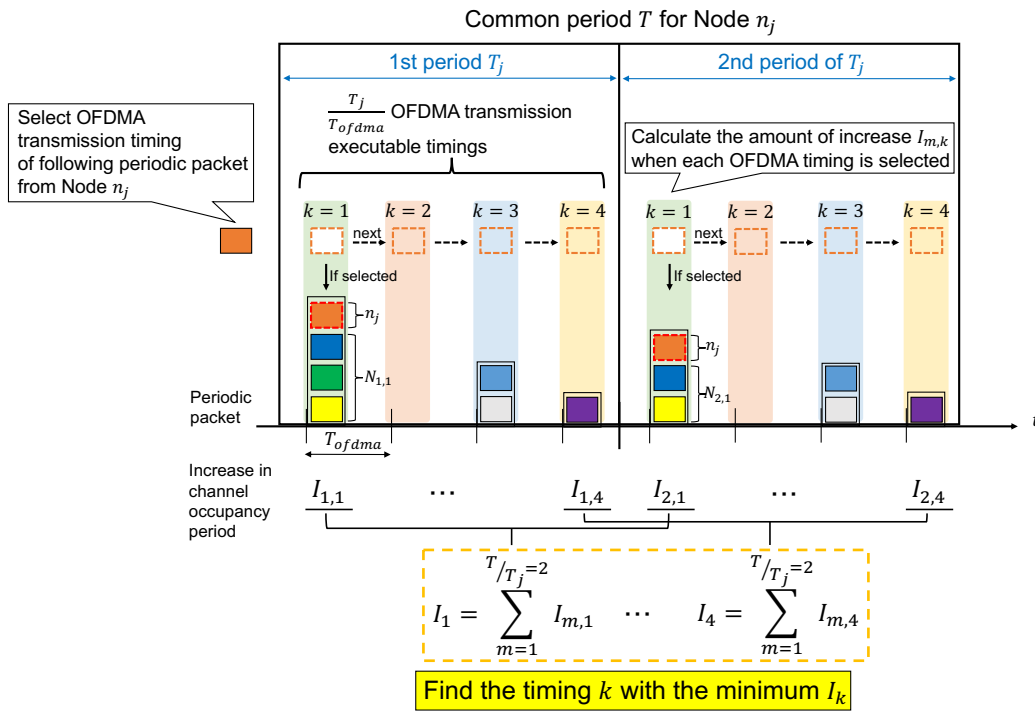


Fig. 1. Overview of the proposed method in case of $T/T_j = 2$ and $T_j/T_{ofdma} = 4$.

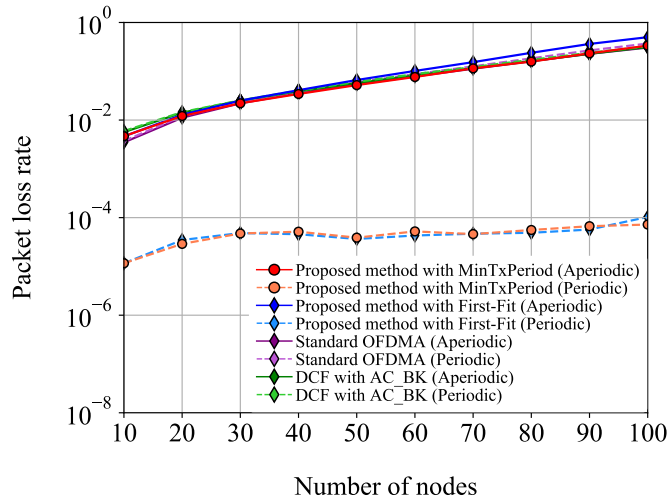


Fig. 2. Packet loss rate.

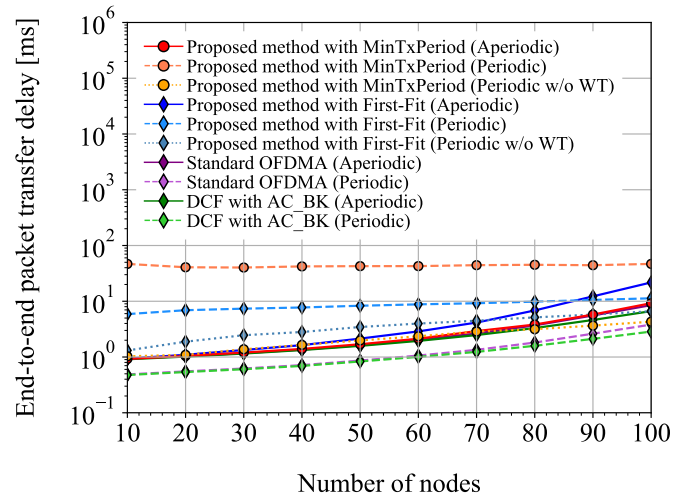


Fig. 3. End-to-end packet transfer delay.

shifting periodic packet generation timings at the applications close to the OFDMA transmission timings, the end-to-end packet transfer delay can be decreased when periodic packets with smaller permissible delay constraints are transferred. Standard OFDMA shows little difference in performance from DCF with AC_BK because it does not provide sufficient OFDMA transmission opportunities based on BSR in this simulation environment where the packet generation rate is low and retransmissions are not considered.

V. CONCLUSION

In this paper, we proposed the bundling method of the periodically generated packets into uplink OFDMA transmissions for collision-free packet transfer as a first step towards constructing a communication infrastructure that achieves high

quality and delay-critical transmission regardless of the packet type. By optimizing channel utilization among periodic packets, the free channel time is increased and the overall efficiency of the wireless channel is improved. In the future, as a next step, we will extend the proposed method to suppress the packet transfer delay for both periodic and aperiodic packets within permissible delays.

REFERENCES

- [1] Wireless Network Research Center, Network Research Institute, NICT, "Introducing manufacturing systems with wireless communications," May 2022. Accessed: Aug. 2023. [Online]. Available: <https://www2.nict.go.jp/wireless/en/ffpj-deploy.html>
- [2] IEEE Std 802.11-1999 (Reaff 2003), Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications, 2003.

- [3] IEEE Std 802.11ax-2021, Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 1: Enhancements for High-Efficiency WLAN, 2021.
- [4] E. Khorov, A. Kiryanov, A. Lyakhov, and G. Bianchi, "A Tutorial on IEEE 802.11ax High Efficiency WLANs," *IEEE Communications Surveys and Tutorials*, vol. 21, no. 1, pp. 197-216, Feb. 2019.
- [5] Q. Chen and Y. -H. Zhu, "Scheduling Channel Access Based on Target Wake Time Mechanism in 802.11ax WLANs," *IEEE Transactions on Wireless Communications*, vol. 20, no. 3, pp. 1529-1543, Mar. 2021.
- [6] Q. Chen, "An Energy-Efficient Channel Access With Target Wake Time Scheduling for Overlapping 802.11ax Basic Service Sets," *IEEE Internet of Things Journal*, vol. 9, no. 19, pp. 18973-18986, Oct. 2022.
- [7] D. Akhmetov, D. Das, D. Cavalcanti, J. Ramirez-Perez, and L. Cariou, "Scheduled Time-Sensitive Transmission Opportunities over Wi-Fi," *Proc. IEEE GLOBECOM 2022*, pp. 1807-1812, Dec. 2022.
- [8] D. Cavalcanti, J. Fang, M. Park, D. Das, D. Akhmetov, and L. Cariou, "Communication of Asynchronous Publication Classification Ultra-Low Latency Transmissions within a Synchronized Transmission Opportunity (S-TXOP)," Intel Corporation, US 20220400503(A1). Jun. 2022.
- [9] B. Schneider et al., "Deterministic Channel Access Using MU EDCA in OFDMA-based Wi-Fi Networks," *Proc. IEEE WFCs 2023*, Apr. 2023.
- [10] QualNet Simulator Version 9.0, Scalable Network Technologies, "<https://www.scalable-networks.com/>".
- [11] IEEE Std 802.11e-2005, Wireless LAN Medium Access Control (MAC) and Physical layer (PHY) specifications: Amendment8: Medium Access Control (MAC) Quality of Service Enhancements, 2005.